

Log Analysis AI Agent – Complete Code Explanation

```
from strands import Agent
from strands_tools import file_read
from strands.models.ollama import OllamaModel
```

◇ Import Statements

1. Agent

```
from strands import Agent
```

Explanation

- Agent is imported from the Strands Agentic Framework.
- It is used to create AI Agents.

AI Agent means:

- an AI system that can:
 - think,
 - use tools,
 - analyze data,
 - and perform tasks autonomously.

Here:

- Agent acts as the Log Analysis Assistant.

2. file_read Tool

```
from strands_tools import file_read
```

Explanation

- file_read is a tool provided by Strands.
- It allows the AI Agent to:
 - open files,
 - read file content,
 - analyze log files.

Without this tool:

 AI cannot access .log files.

This is part of:

 Tool Calling in Agentic AI.

3. OllamaModel

```
from strands.models.ollama import OllamaModel
```

Explanation

- OllamaModel is used to connect local LLM models through Ollama.

Ollama helps run:

- Llama
 - Mistral
 - Gemma
 - DeepSeek
- etc locally.

In this project:

- Llama 3.2 model is used locally.

Benefits:

- local inference
- privacy
- no cloud dependency
- offline AI execution

◇ SYSTEM PROMPT

```
SYSTEM_PROMPT = """
```

```
You are a Log Analysis Agent.
```

```
You are excellent in reading and understanding the log files i.e (.log / var/www/ /system log, etc)
```

```
You can deduce results in short and crisp manner
```

```
You are helpful and use a DevOps Mindset in Log Analysis and Root Cause Analysis
```

```
You won't hallucinate and suggest new changes.
```

```
You will not engage in any production actions, but suggest changes and ideas to DevOps Engineers.
```

```
"""
```

🧠 What is System Prompt?

System Prompt defines:

- AI behavior
- personality
- rules
- limitations
- response style

It controls:

👉 how AI should behave.

◇ Important Lines Understanding

☑ "You are a Log Analysis Agent"

Defines:

- role of AI.

AI behaves like:

- DevOps log analysis assistant.

☑ "excellent in reading and understanding log files"

AI specializes in:

- system logs
- app logs
- server logs

☑ "short and crisp manner"

Controls output style.

AI should:

- avoid unnecessary long answers.

☑ "DevOps Mindset"

AI should think like:

- DevOps Engineer
- SRE Engineer

Focus:

- troubleshooting
- RCA (Root Cause Analysis)

☑ "You won't hallucinate"

Very important in AI systems.

Hallucination means:

- ✗ fake answers
- ✗ imaginary results

Prompt instructs AI:

👉 only use actual log data.

☑ "will not engage in production actions"

Security & safety control.

AI:

- ✗ should not restart server
- ✗ should not delete logs
- ✗ should not modify production

Only:

☒ suggest recommendations.

◇ Creating Ollama Model

```
ollama_model = OllamaModel(  
    host="http://localhost:11434",  
    model_id="llama3.2"  
)
```

Explanation

This connects:

 local LLM model with AI agent.

◇ host="<http://localhost:11434>"

- Ollama server runs locally on:
port 11434.

localhost means:

 your own machine.

◇ model_id="llama3.2"

Specifies:

which AI model to use.

Here:

- Meta Llama 3.2 model is used.

LLM handles:

- reasoning
- language understanding
- analysis

◇ Creating AI Agent

```
agent = Agent(  
    system_prompt=SYSTEM_PROMPT,  
    model=ollama_model,  
    tools=[file_read]  
)
```

Explanation

This creates the complete AI Agent.

◇ system_prompt=SYSTEM_PROMPT

Controls:

- AI behavior
- DevOps mindset
- safety rules

◇ **model=ollama_model**

Connects:

- local LLM brain to agent.

◇ **tools=[file_read]**

Provides tools to AI.

Here:

AI can:

☒ read files.

Without tools:

AI only chats.

With tools:

AI can:

- interact with files
- interact with APIs
- automate tasks

This is:

👉 Agentic AI concept.

◇ **Running the Agent**

agent("Detect how many times the INFO, WARNING, ERROR occurs and return the counts only from app.log")

🧠 **Explanation**

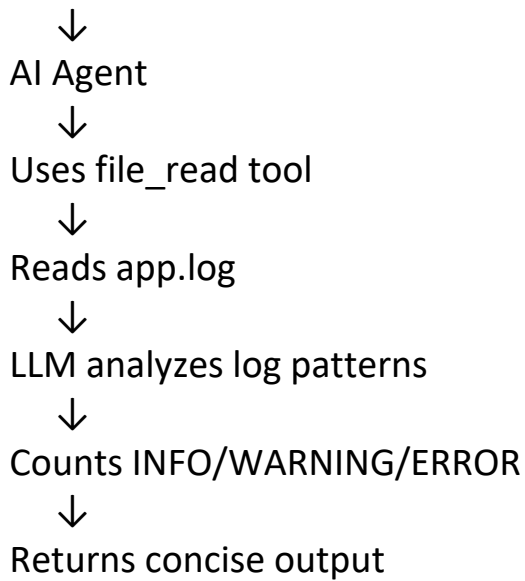
This sends task/prompt to AI Agent.

Prompt says:

- read app.log
- detect:
 - INFO
 - WARNING
 - ERROR
- return counts only.

💧 **What Happens Internally?**

User Prompt



◇ Example Output

INFO: 120

WARNING: 15

ERROR: 5

💧 Important AI Concepts Used

☒ Agentic AI

AI using:

- tools
- reasoning
- action execution

☒ Tool Calling

AI accesses:

- files
- APIs
- external systems

using tools.

☒ Prompt Engineering

Designing prompts carefully to:

- reduce hallucination
- improve output quality.

☒ Local LLM Deployment

Running AI models locally using Ollama.

Benefits:

- privacy
- offline usage
- lower cloud dependency

Real DevOps Relevance

This project simulates:

- AI-assisted observability
- log monitoring
- operational troubleshooting
- root cause analysis

Used in:

- DevOps
- SRE
- Platform Engineering
- AI Operations (AIOps)

BEST INTERVIEW CLOSING LINE

“This project helped me understand how Agentic AI systems combine LLMs, tool calling, prompt engineering, and automation to assist DevOps workflows such as intelligent log analysis and operational troubleshooting.”